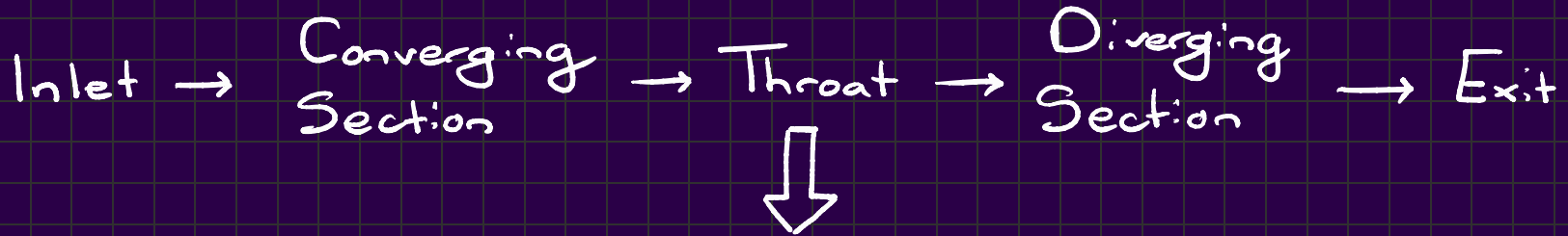


Reading Nozzle - Axis Plots



$$M_t = 1$$

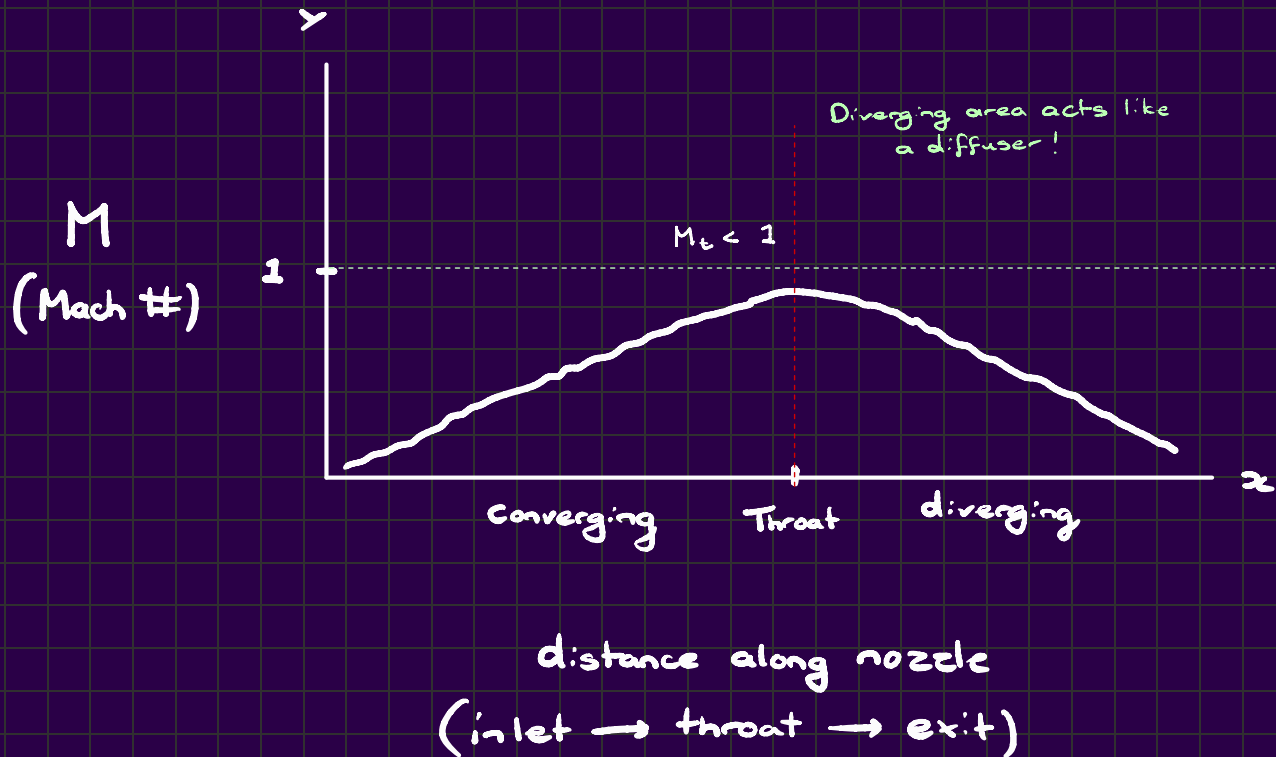
① Static Pressure Plot

② Mach # Plot

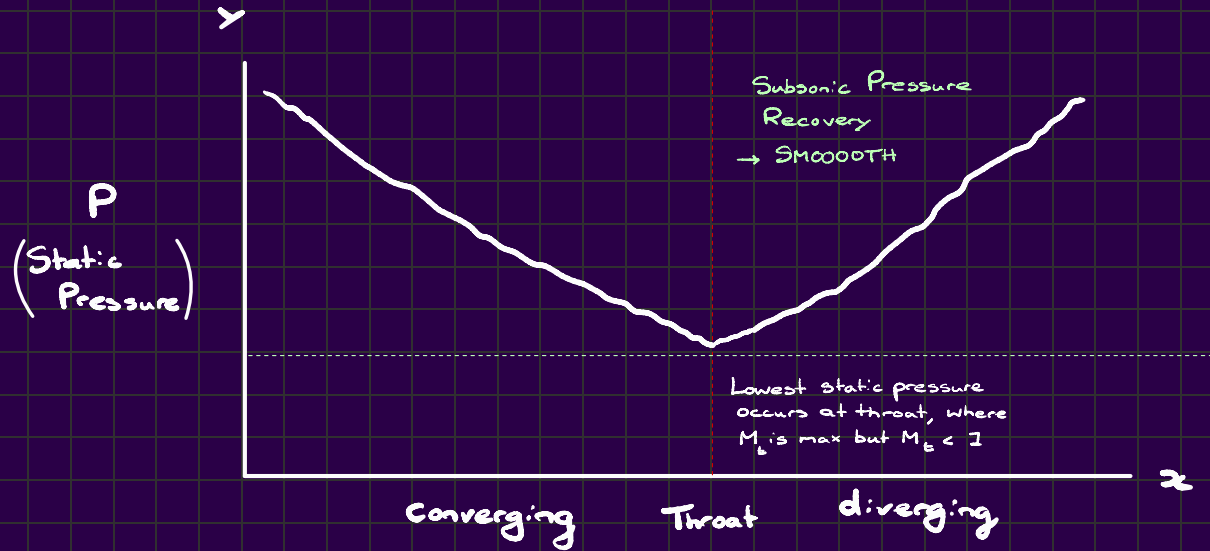
③ Stagnation Pressure Plot

Fully Subsonic Pressure & Mach Sketch

$M(x)$ (Mach Sketch)

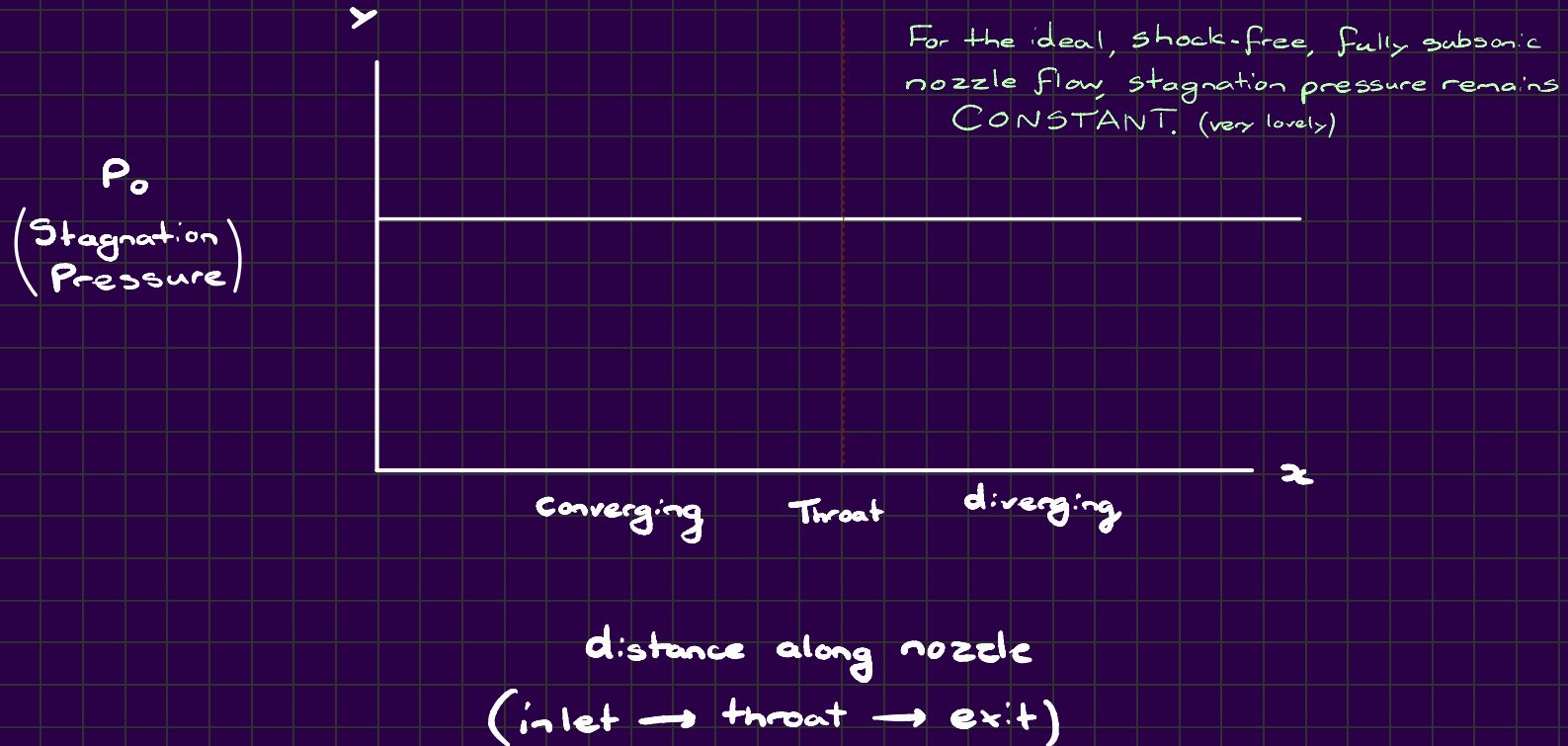


$P(x)$ (Static Pressure Sketch)



distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)

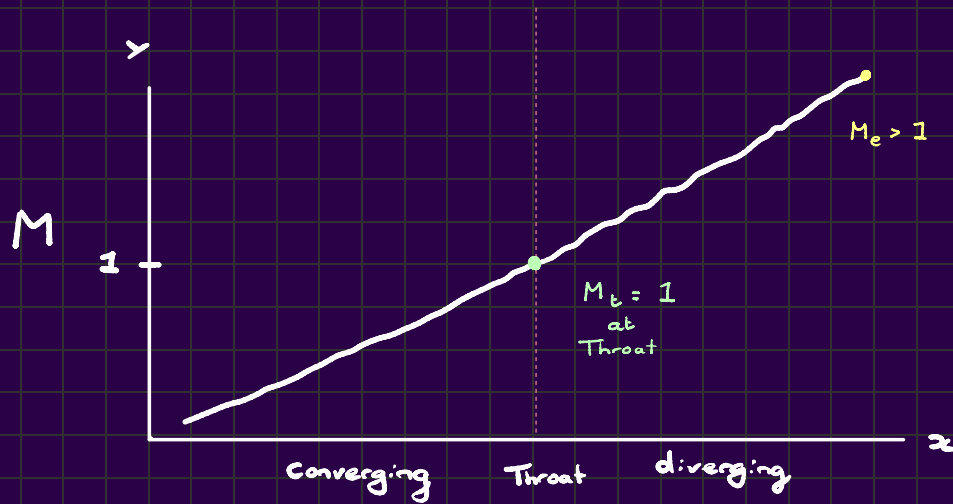
$P_o(x)$ (Stagnation Pressure Sketch)



Isentropic Supersonic/Design Sketch

The flow: subsonic $\rightarrow$ sonic at throat $\rightarrow$ supersonic

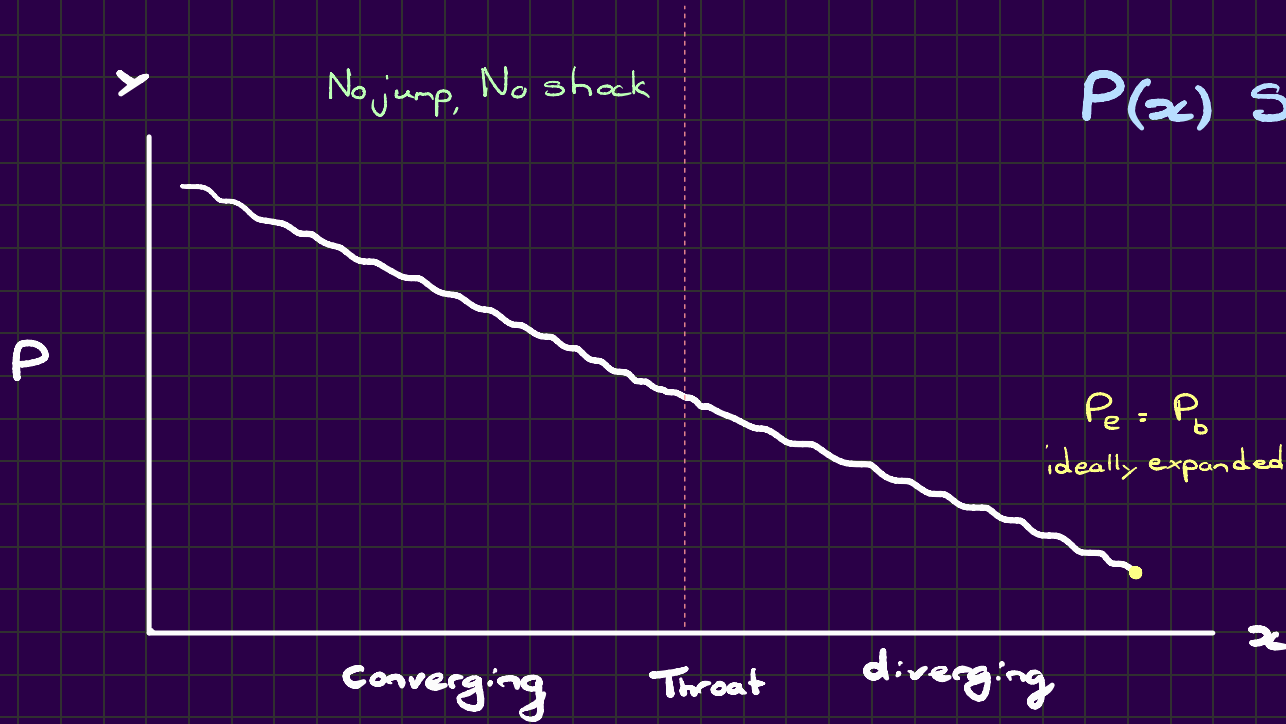
Condition: $M_b = 1$, $M_e > 1$, $P_e = P_b$



$M(x)$ Sketch

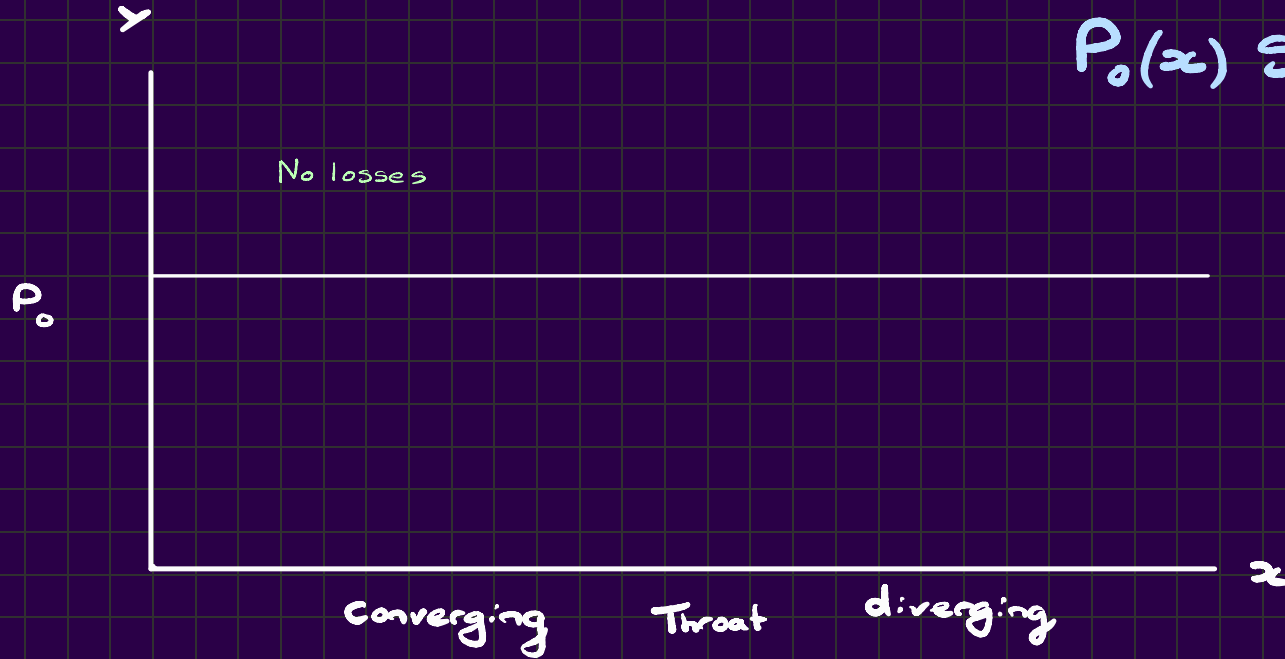
distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)

$P(x)$ Sketch



distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)

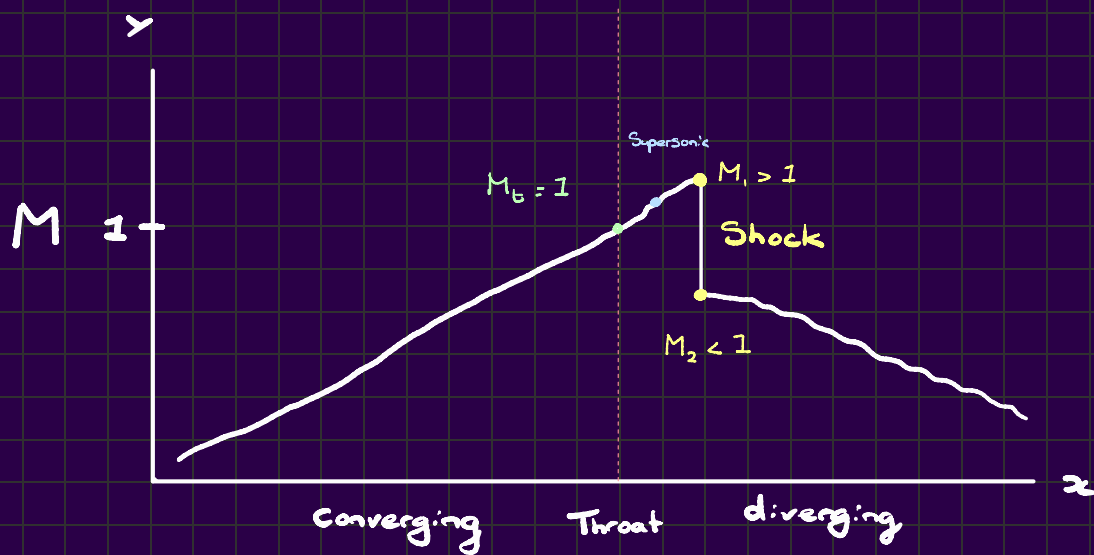
$P_0(x)$ Sketch



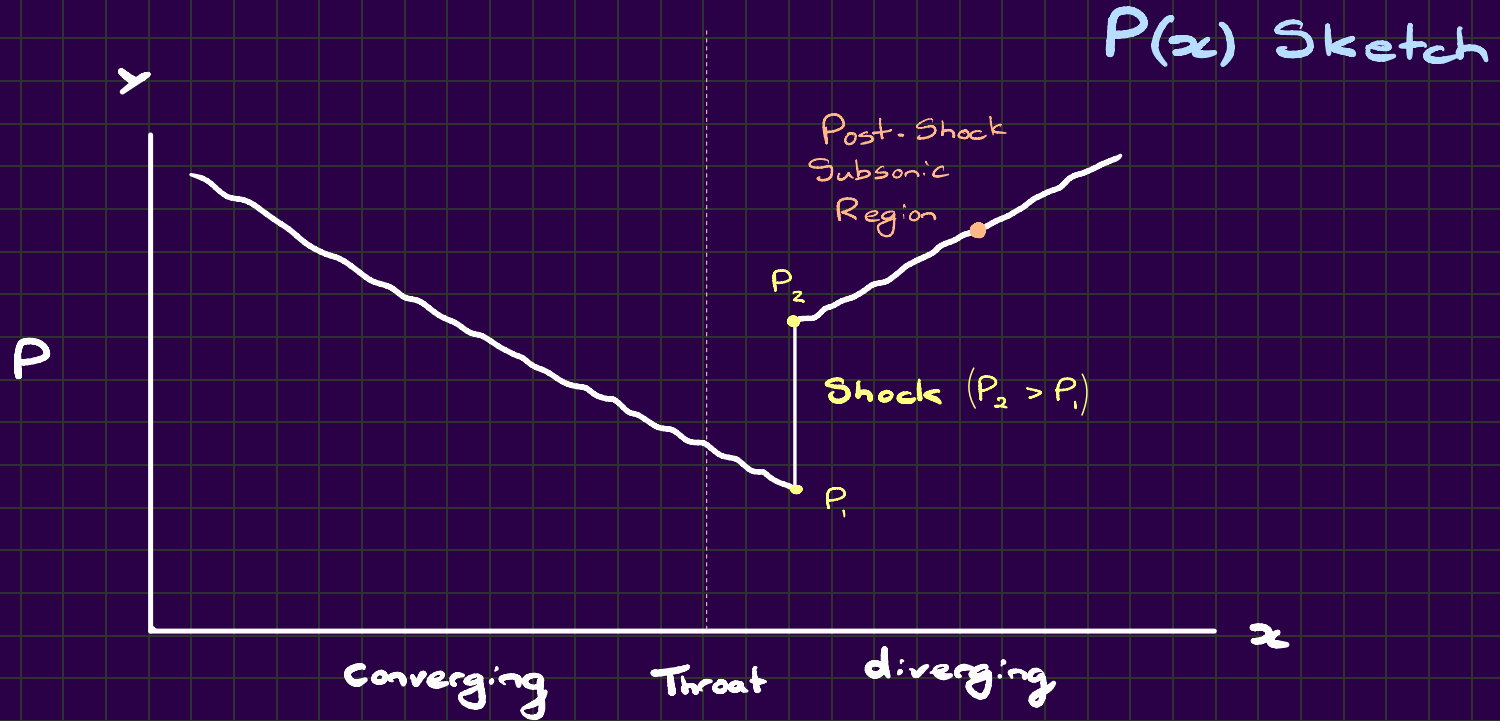
distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)

Internal Normal Shock Sketch

$M(x)$ Sketch

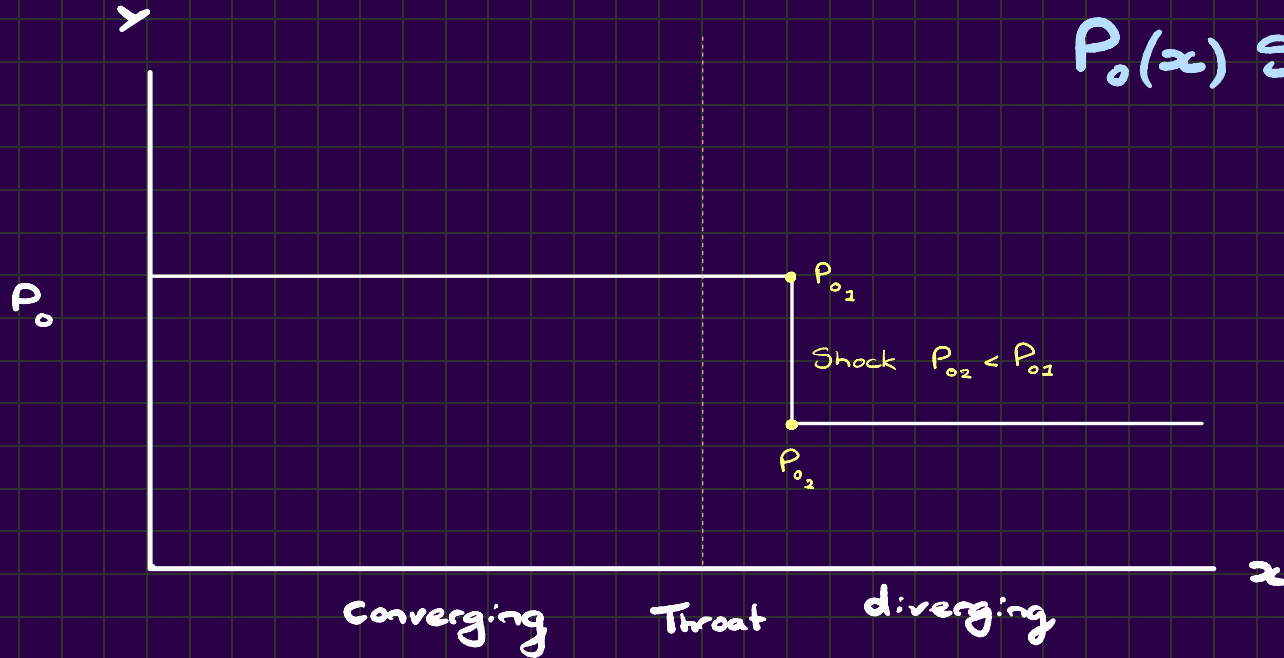


distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)



distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)

$P_o(x)$ Sketch



distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)

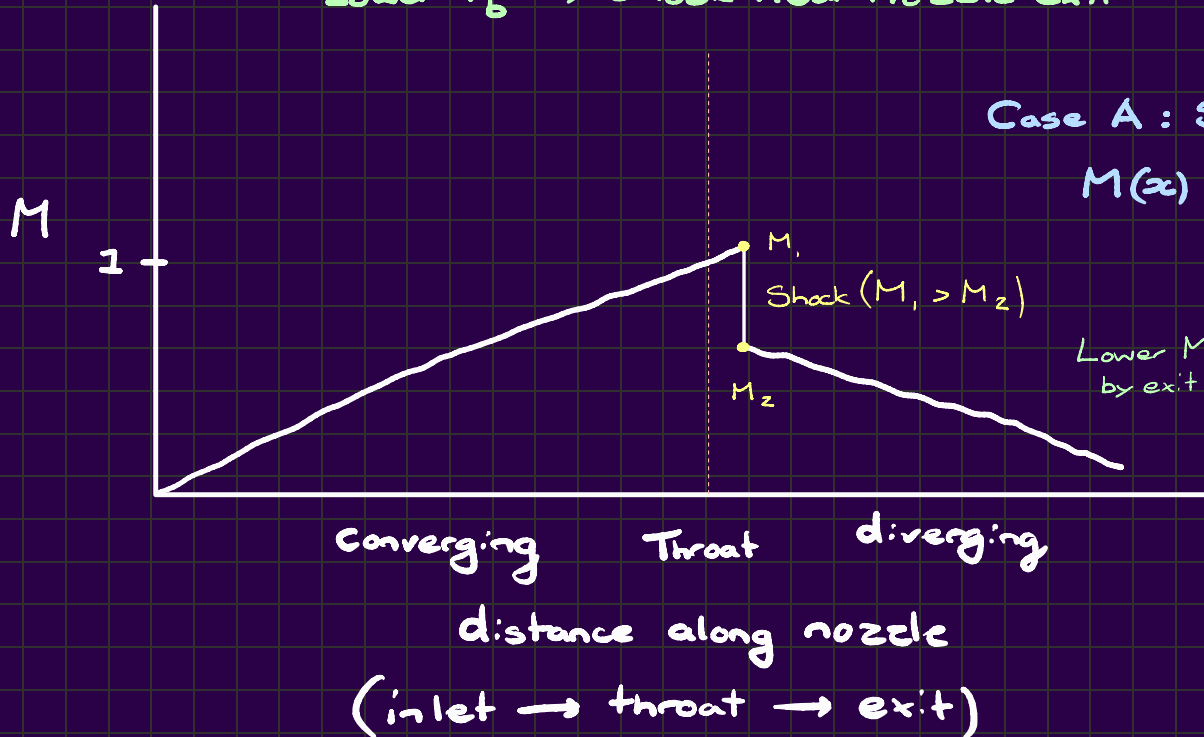
Shock Movement Sketches

Higher $P_b \rightarrow$ Shock near throat

Lower $P_b \rightarrow$ Shock near nozzle exit

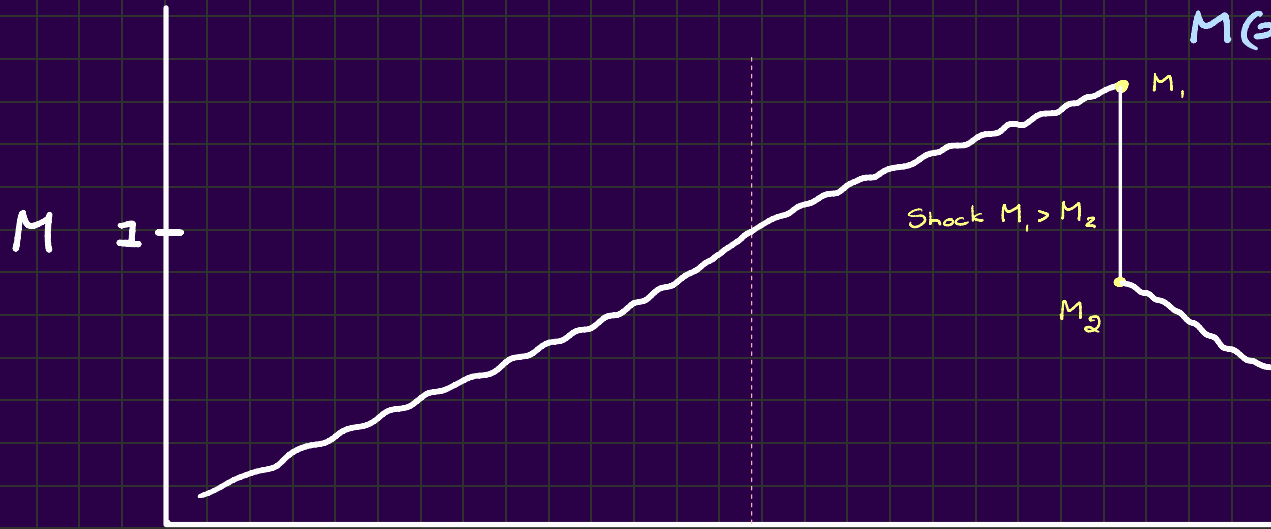
Case A : Shock near Throat

$M(x)$ Sketch



Case B : Shock near Exit

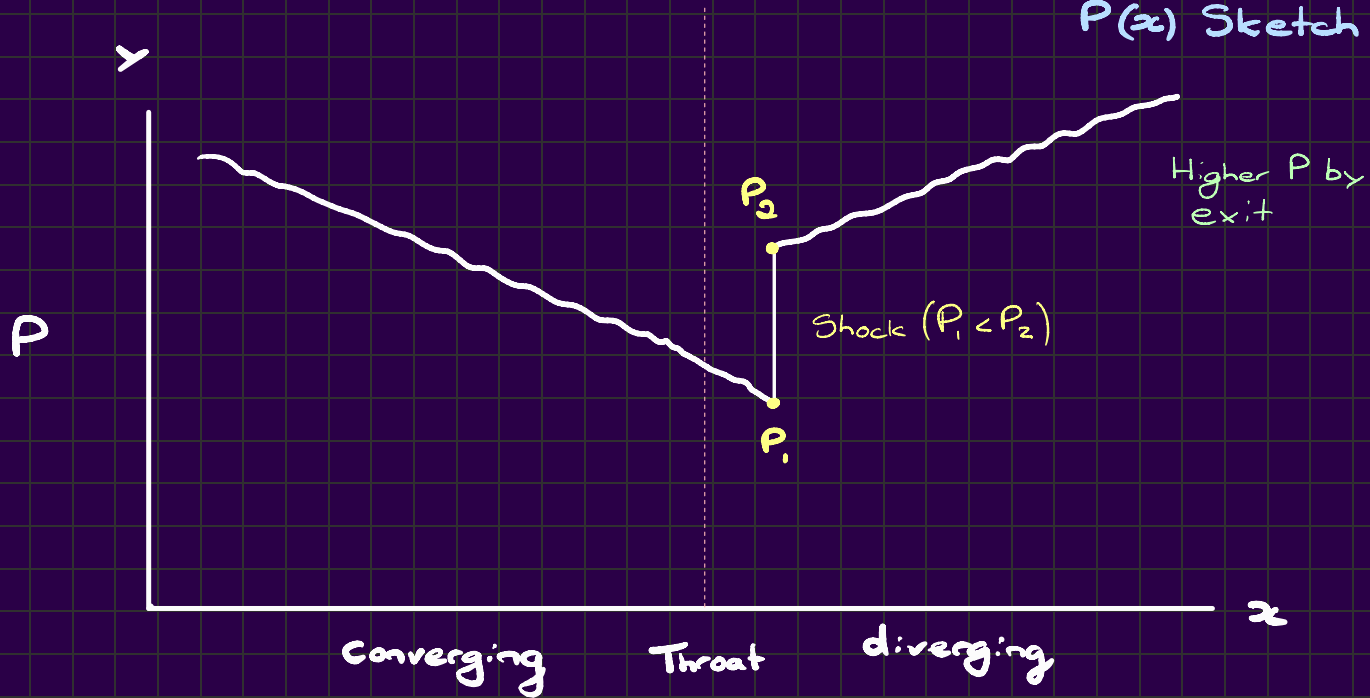
$M(x)$ Sketch



converging Throat diverging
distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)

Case A : Shock near Throat

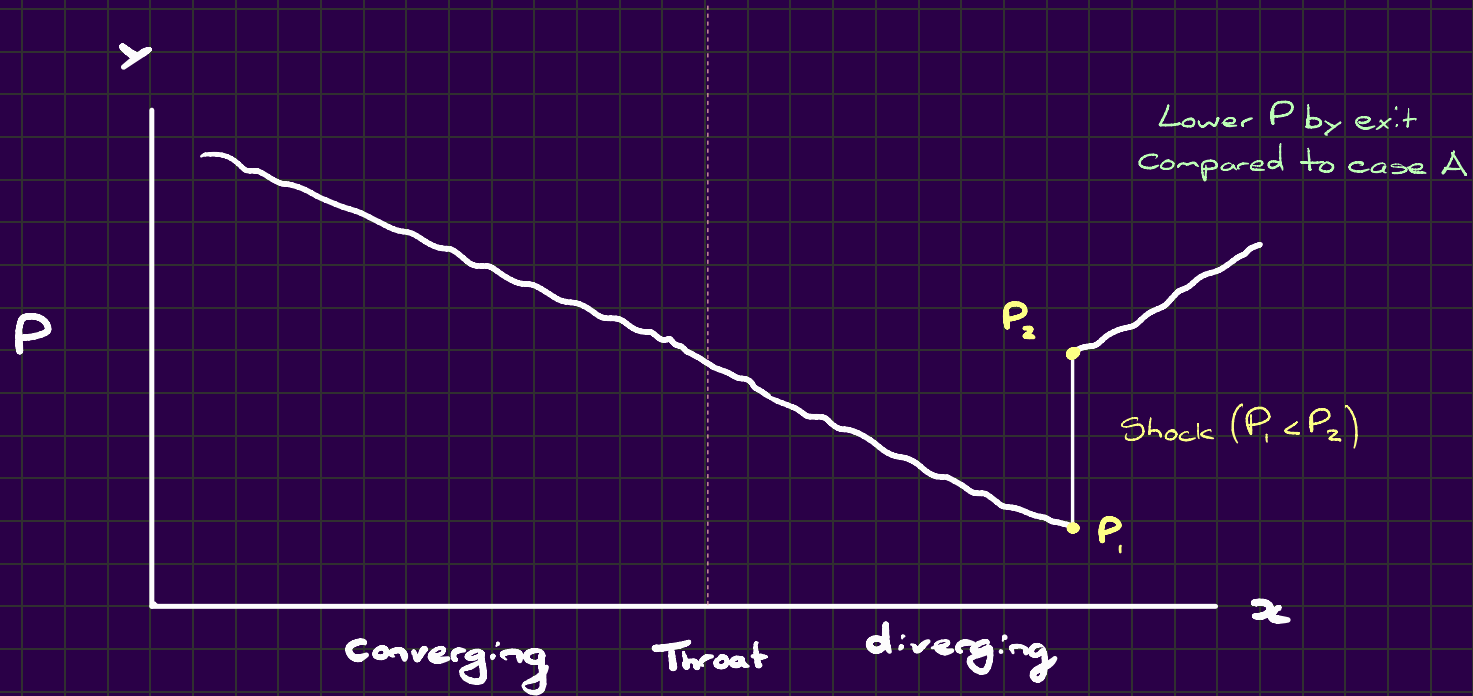
$P(x)$ Sketch



distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)

Case B : Shock near Exit

$P(x)$ Sketch

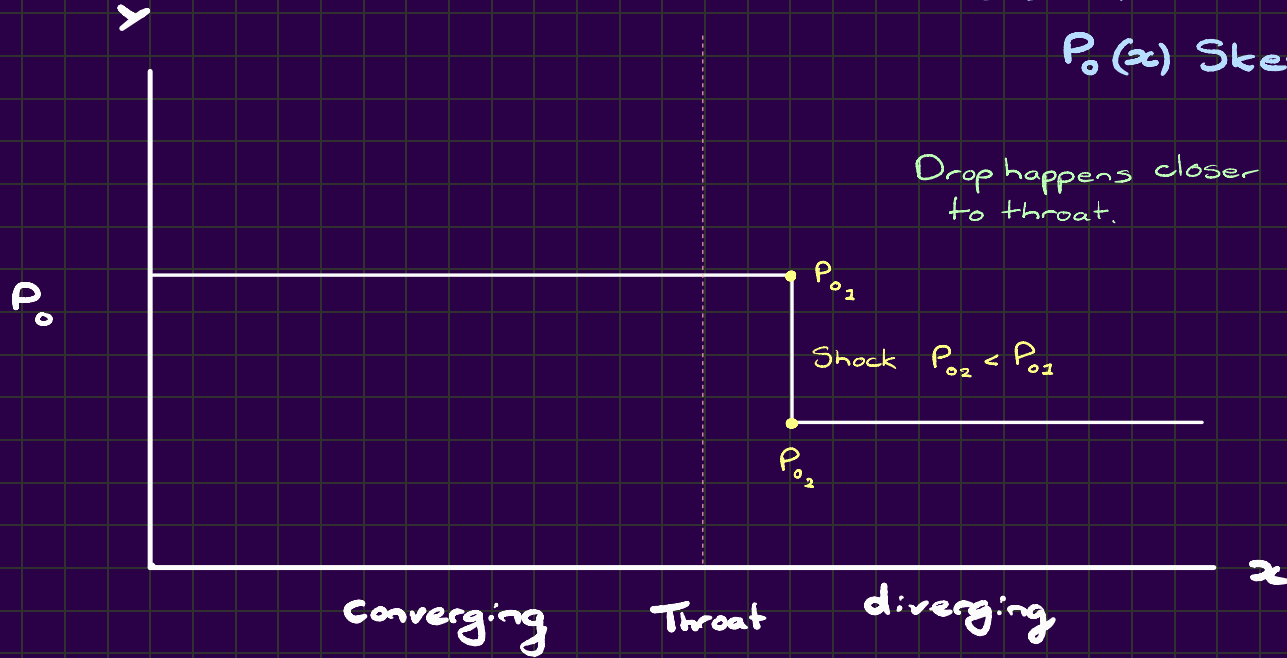


distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)

Case A : Shock near Throat

$P_o(x)$ Sketch

Drop happens closer
to throat.



distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)

Case B : Shock near Exit

$P_0(x)$ Sketch

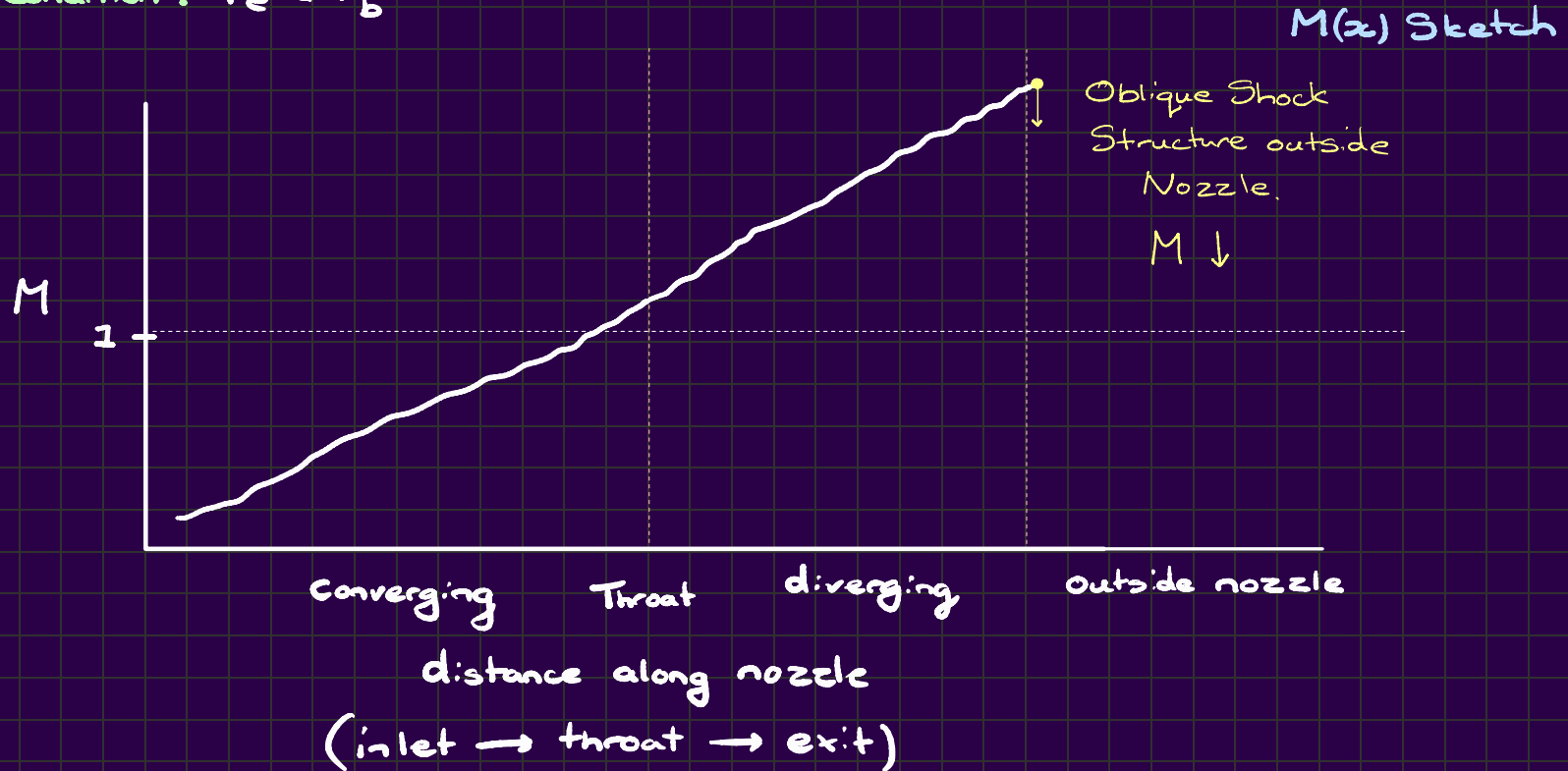


distance along nozzle
(inlet $\rightarrow$ throat $\rightarrow$ exit)

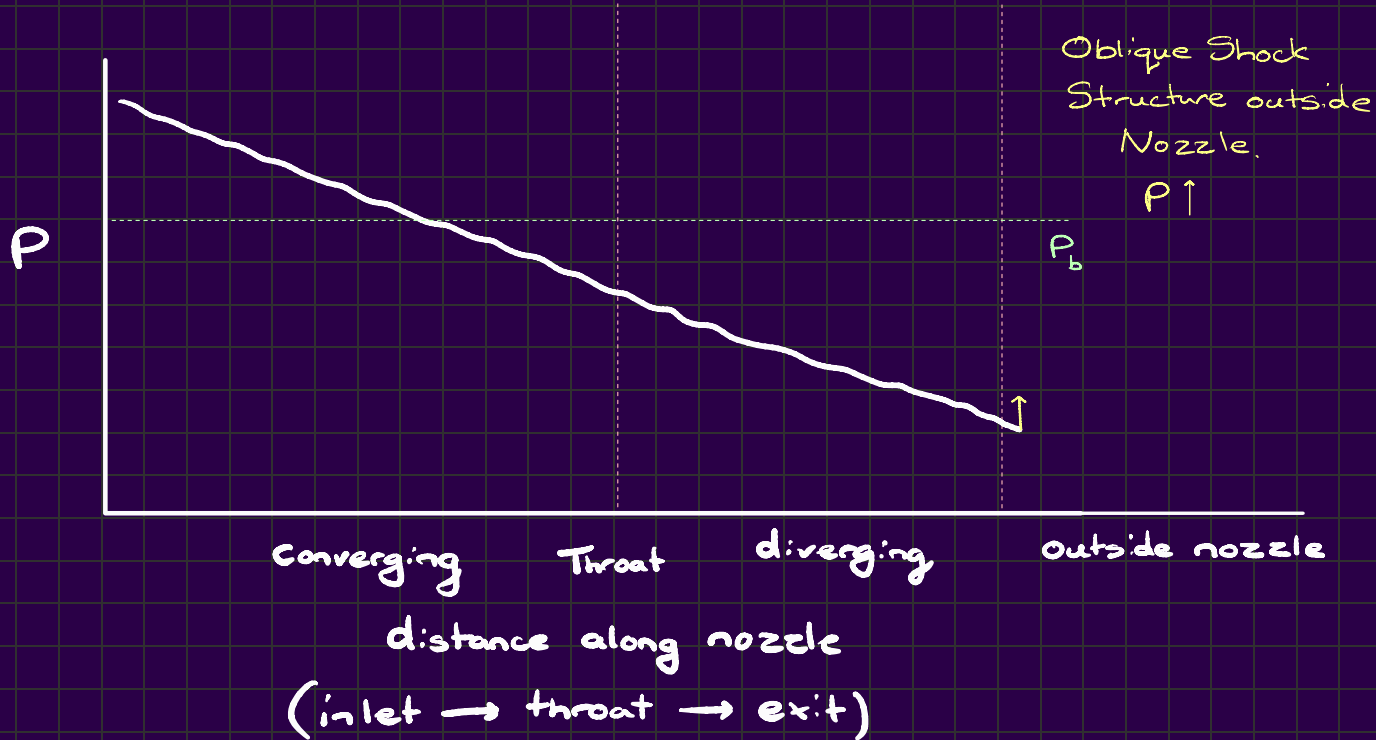
Over - Expanded Flow Sketch

The flow : Subsonic $\rightarrow$ Sonic at Throat $\rightarrow$ Supersonic to Exit

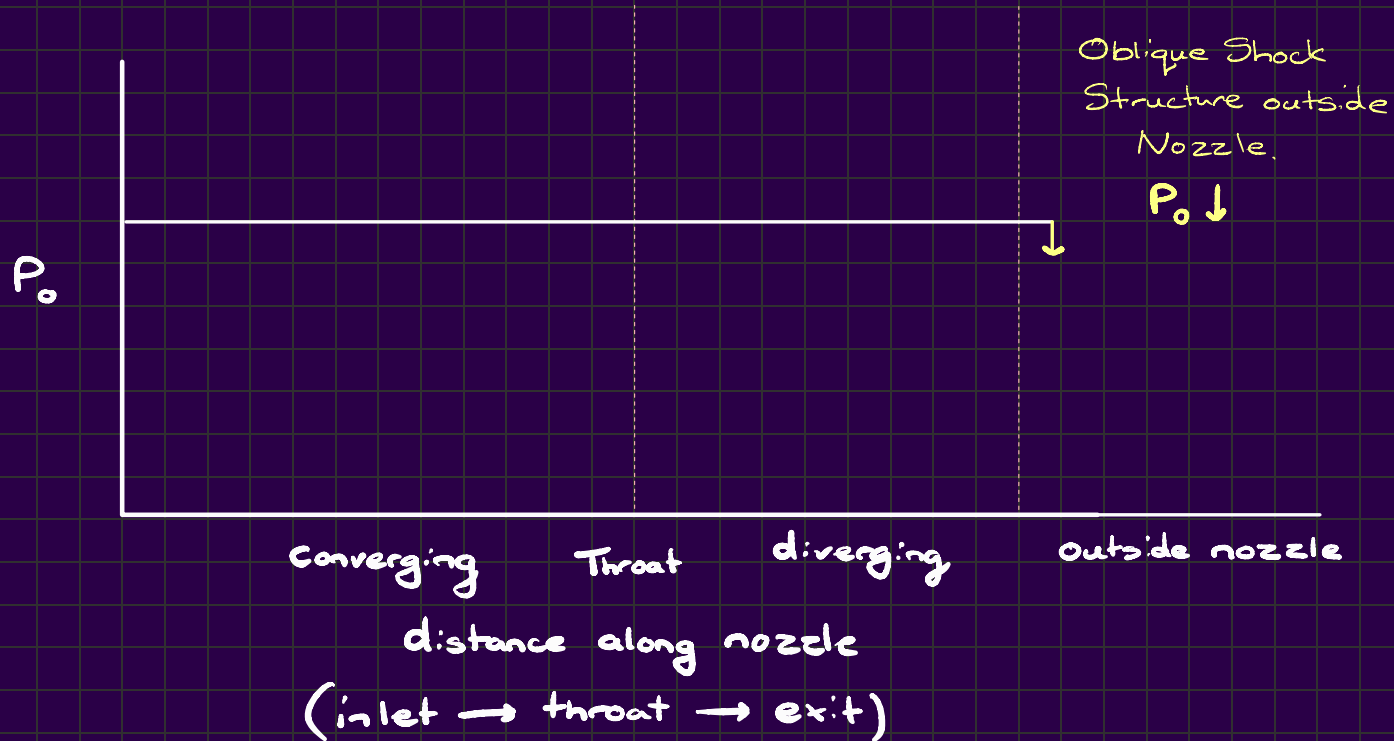
Condition : $P_e < P_b$



$P(x)$ Sketch



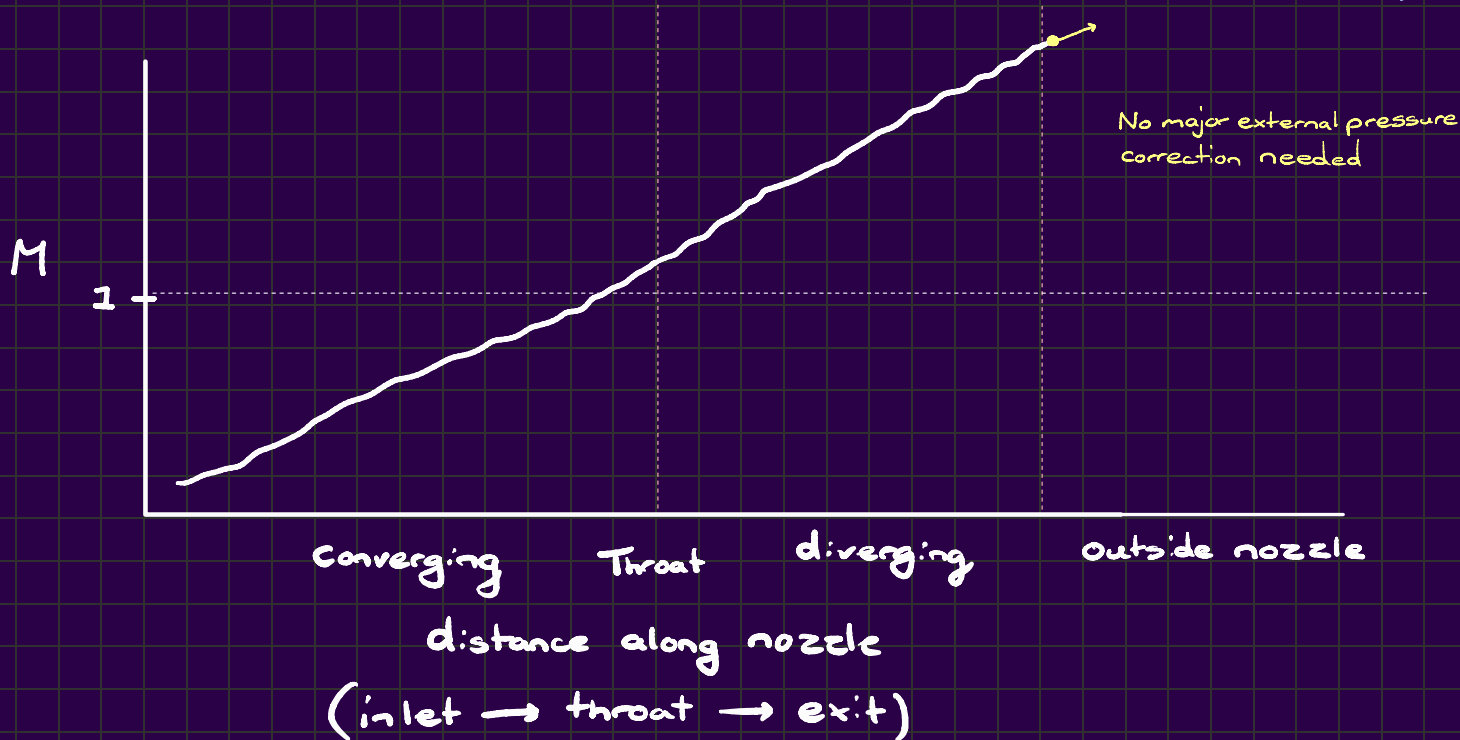
$P_0(x)$ Sketch



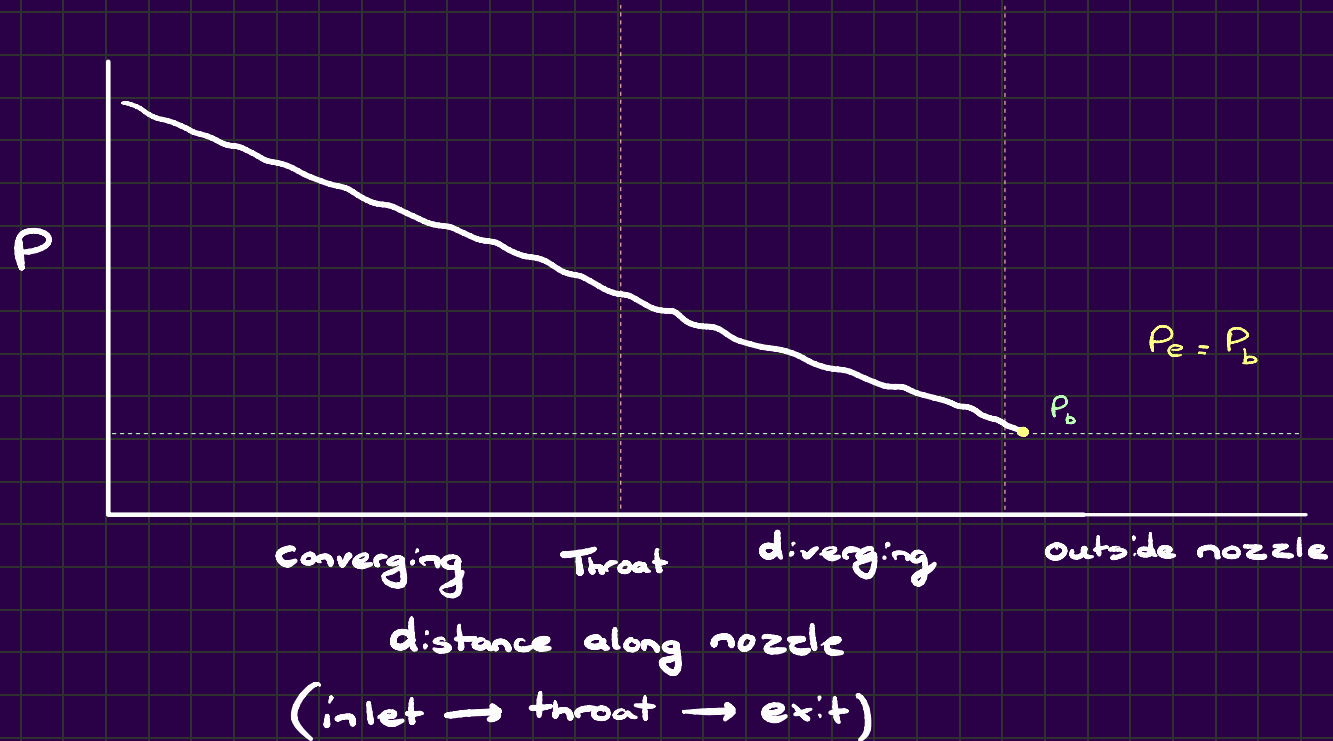
Ideally Expanded Flow Sketch

Condition : $P_e = P_b$

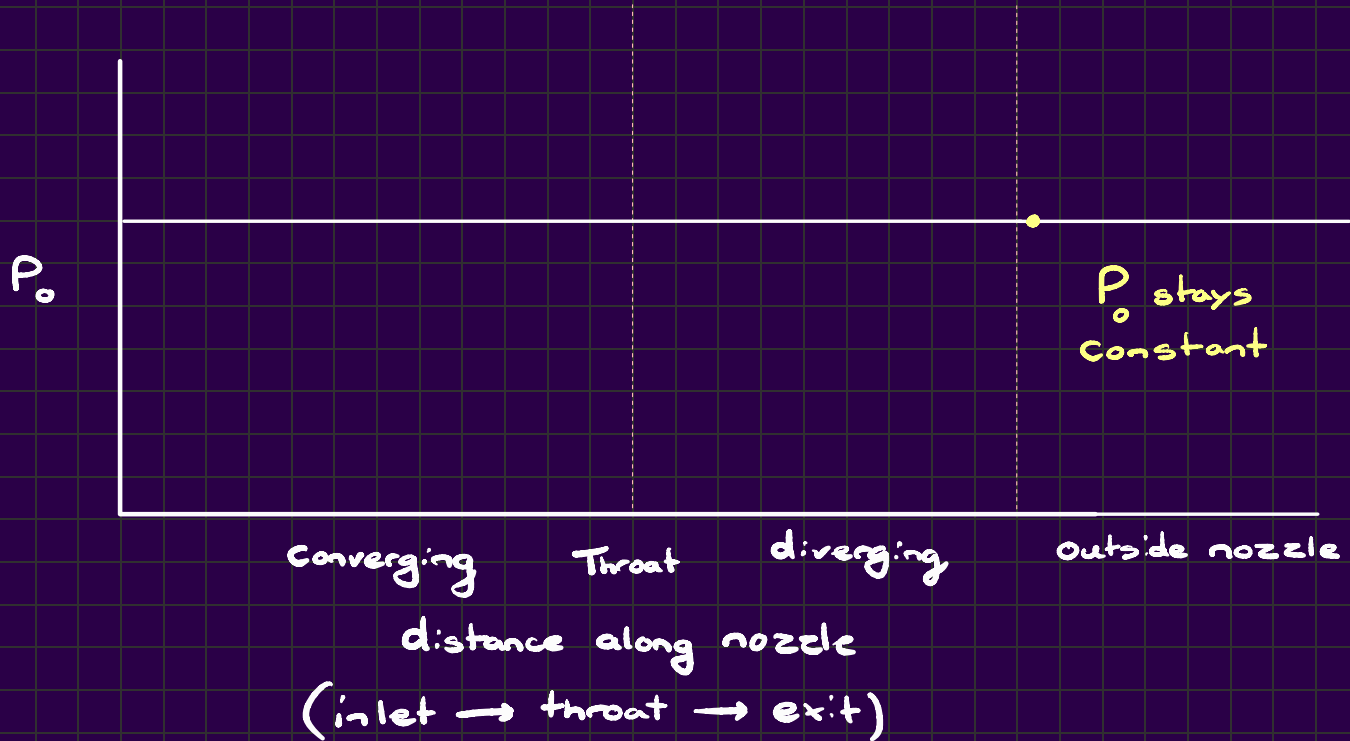
$M(x)$ Sketch



$P(x)$ Sketch



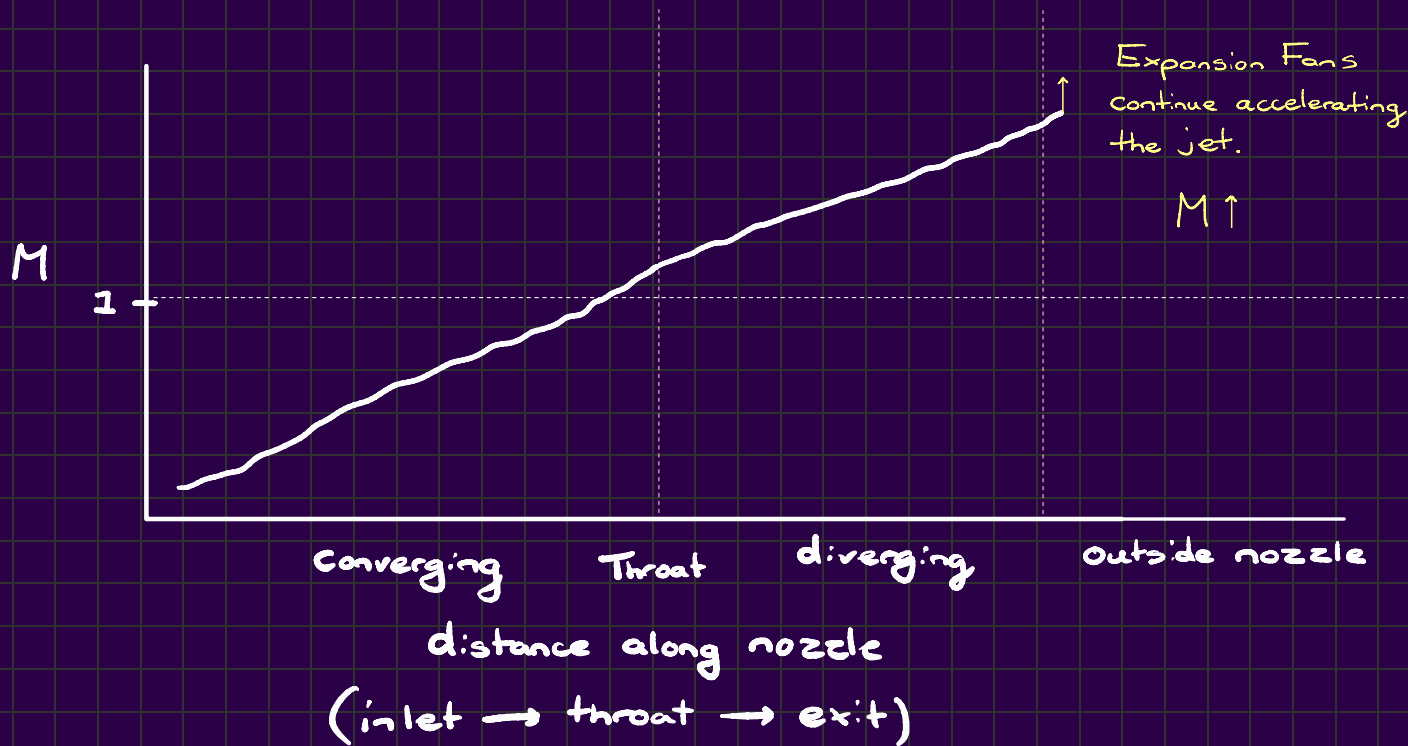
$P_o(\infty)$ Sketch



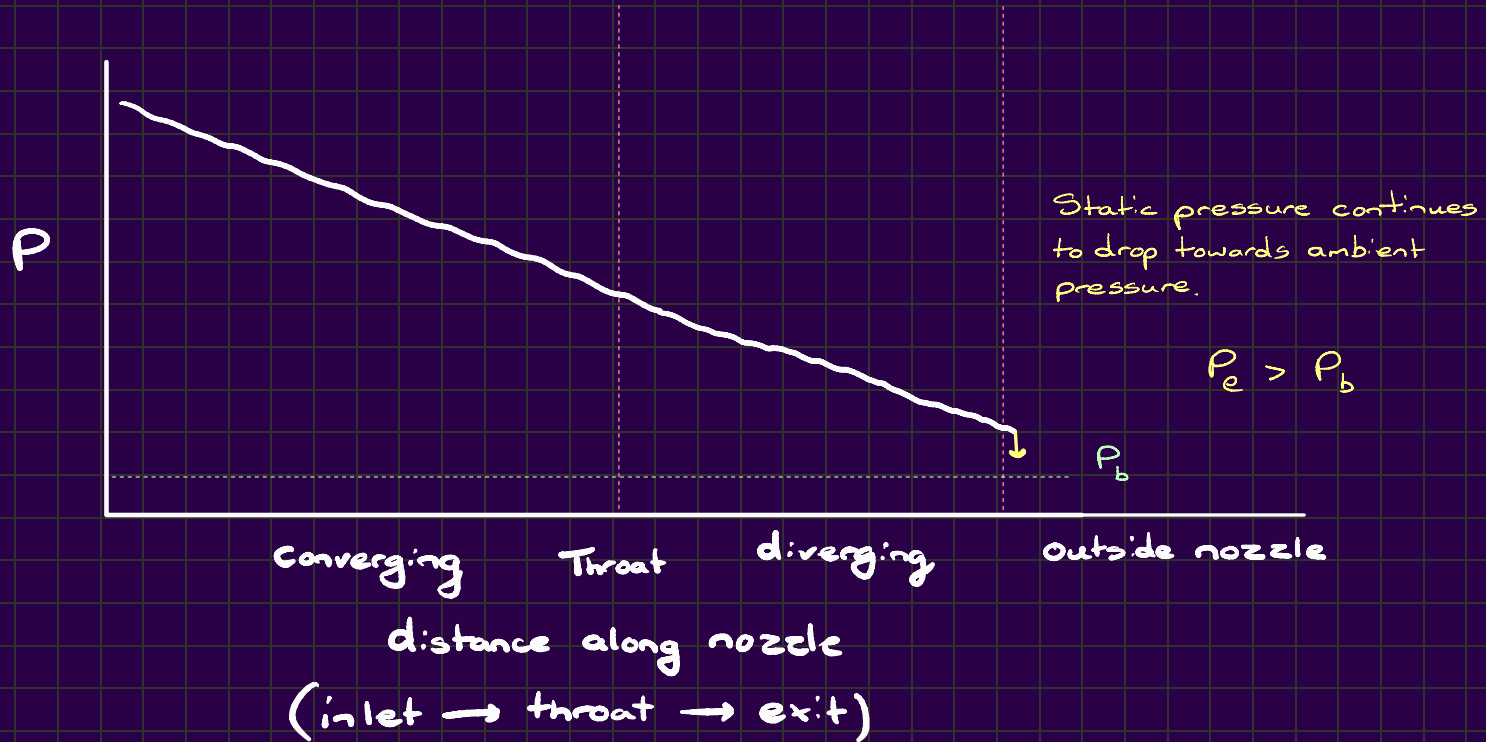
Under-Expanded Flow Sketch

Flow: $P_e > P_b$

$M(x)$ Sketch



$P(x)$ Sketch



$P_o(\infty)$ Sketch

